

MOOSE JAW, SK, Canada

WMO: 715390

Lat: 50.332N

Lon: 105.537W

Elev: 577

StdP: 94.58

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-28.0	-34.9	0.1	-30.7	-31.7	0.2	-28.0	14.4	-3.7	13.2	-5.7	3.8	280	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	31.8	18.6	29.7	18.0	27.7	17.3	21.1	27.4	19.8	26.3	18.7	25.1	5.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.9	23.7	17.6	13.5	22.6	16.3	12.5	21.4	64.1	27.5	59.3	26.2	55.3	25.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.5	11.1	9.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-35.1	35.5	3.1	2.2	-37.3	37.1	-39.2	38.4	-40.9	39.6	-43.1	41.3
			-35.2	23.6	3.1	1.6	-37.4	24.7	-39.2	25.6	-41.0	26.5	-43.2	27.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-12.2	-10.9	-4.0	4.5	11.1	16.2	19.1	18.2	13.0	5.2	-3.3	-9.9	(6)
(7)		DBStd	12.81	8.82	8.26	8.09	5.74	4.79	3.64	3.05	3.55	4.62	5.16	7.02	8.29	(7)
(8)		HDD10.0	3131	688	584	434	177	44	2	0	0	22	163	398	618	(8)
(9)		HDD18.3	5354	946	818	692	415	228	84	27	47	167	407	648	877	(9)
(10)		CDD10.0	941	0	0	1	12	78	187	282	253	113	14	1	0	(10)
(11)		CDD18.3	123	0	0	0	0	4	19	51	42	8	0	0	0	(11)
(12)		CDH23.3	1913	0	0	0	8	104	266	655	662	210	9	0	0	(12)
(13)		CDH26.7	641	0	0	0	1	23	70	221	256	70	1	0	0	(13)
(14)	Wind	WSAvg	4.8	5.1	4.9	5.1	5.1	4.6	4.1	4.0	4.4	4.8	5.0	5.0	(14)	
(15)	Precipitation	PrecAvg	375	19	16	18	26	50	63	53	36	37	19	16	20	(15)
(16)		PrecMax	510	38	33	51	76	129	169	132	119	100	94	41	53	(16)
(17)		PrecMin	224	3	2	0	1	1	14	20	3	2	0	1	6	(17)
(18)		PrecStd	78	9	9	11	21	31	39	32	29	27	20	11	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	7.7	16.6	24.2	29.3	32.2	34.2	35.2	31.7	24.4	16.6	8.3	(19)
(20)		MCWB	3.3	4.3	8.6	11.9	14.9	18.0	20.3	18.7	16.7	12.9	8.8	3.7	(20)	
(21)		2%	DB	4.4	4.5	12.1	20.5	26.2	28.5	31.4	32.0	28.9	19.9	11.6	5.8	(21)
(22)		MCWB	1.9	2.0	6.5	10.4	13.8	17.6	20.1	18.2	15.7	11.2	6.0	2.4	(22)	
(23)		5%	DB	2.7	2.5	8.9	17.7	23.6	26.3	29.1	29.6	25.6	16.5	8.8	3.7	(23)
(24)		MCWB	0.8	0.8	4.8	9.1	12.6	16.3	19.3	17.7	14.8	9.5	4.5	1.3	(24)	
(25)		10%	DB	0.6	0.6	5.9	14.7	21.1	24.2	27.0	27.1	22.5	13.7	5.9	1.3	(25)
(26)		MCWB	-0.9	-0.5	3.1	7.5	11.4	15.5	18.4	17.1	13.4	8.1	2.8	-0.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	4.3	9.1	13.1	16.7	21.5	24.0	21.9	18.7	14.1	9.0	4.3	(27)
(28)		MDCB	6.3	7.5	15.6	21.4	25.0	28.1	29.8	28.6	27.0	22.4	16.4	7.3	(28)	
(29)		2%	WB	2.1	2.2	6.8	11.1	14.9	19.2	21.9	20.3	16.9	11.7	6.5	2.7	(29)
(30)		MDCB	4.2	4.0	11.3	19.1	23.4	25.5	27.5	27.5	24.9	18.6	11.0	5.4	(30)	
(31)		5%	WB	1.0	0.9	4.9	9.6	13.7	17.6	20.4	19.1	15.5	10.1	4.7	1.3	(31)
(32)		MDCB	2.7	2.3	8.4	16.4	21.4	23.7	26.4	25.9	23.9	15.5	8.2	3.6	(32)	
(33)		10%	WB	-0.9	-0.5	3.2	8.0	12.5	16.4	19.2	18.0	14.0	8.6	3.0	-0.4	(33)
(34)		MDCB	0.5	0.7	5.7	13.8	19.5	22.2	25.4	25.5	21.6	13.2	5.5	1.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.1	10.2	10.4	13.0	14.0	12.3	13.6	14.7	14.1	12.0	10.0	9.4	(35)	
(36)		5% DB	MCDBR	10.7	10.9	14.2	18.3	18.0	16.0	17.4	19.5	20.0	17.5	14.0	11.0	(36)
(37)		MCWBR	8.8	8.8	9.2	9.8	8.7	7.4	8.1	8.1	9.2	9.9	9.1	8.3	(37)	
(38)		5% WB	MCDBR	10.4	10.4	13.5	16.9	15.8	13.8	15.0	16.0	17.8	15.7	13.1	10.8	(38)
(39)		MCWBR	8.8	8.5	9.0	9.6	8.1	7.1	7.9	7.8	8.9	9.3	8.8	8.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.270	0.279	0.332	0.352	0.350	0.363	0.370	0.329	0.286	0.255	0.231	(40)	
(41)		taud	2.358	2.367	2.494	2.378	2.370	2.426	2.400	2.361	2.449	2.547	2.511	2.447	(41)	
(42)		Ebn at Noon	799	881	941	912	904	907	887	862	868	859	811	796	(42)	
(43)		Edh at Noon	63	81	86	109	115	110	111	110	90	68	54	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.27	2.33	3.67	4.82	5.82	6.09	6.58	5.23	3.89	2.33	1.28	0.91	(44)	
(45)		RadStd	0.16	0.20	0.30	0.37	0.48	0.40	0.39	0.35	0.34	0.29	0.18	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.70	N/A	N/A	N/A	N/A

Nomenclature: See separate page